

# On plane spanning trees and cycles of multicolored point sets with few intersections

M. Kano<sup>a</sup>, C. Merino<sup>b,\*</sup>, J. Urrutia<sup>b,2</sup>

<sup>a</sup> Department of Computer and Information Science, Ibaraki University, Hitachi 316-8511, Japan

<sup>b</sup> Instituto de Matemáticas, U.N.A.M. Área de la investigación científica, Circuito Exterior, Ciudad Universitaria Coyoacán 04510, México D.F., Mexico

Received 24 February 2004

Available online 5 January 2005

Communicated by F. Dehne

## Abstract

Let  $P_1, \dots, P_k$  be a collection of disjoint point sets in  $\mathbb{R}^2$  in general position. We prove that for each  $1 \leq i \leq k$  we can find a plane spanning tree  $T_i$  of  $P_i$  such that the edges of  $T_1, \dots, T_k$  intersect at most  $(k-1)(n-k) + \frac{1}{2}k(k-1)$ , where  $n$  is the number of points in  $P_1 \cup \dots \cup P_k$ . If the intersection of the convex hulls of  $P_1, \dots, P_k$  is nonempty, we can find  $k$  spanning cycles such that their edges intersect at most  $(k-1)n$  times, this bound is tight. We also prove that if  $P$  and  $Q$  are disjoint point sets in general position, then the minimum weight spanning trees of  $P$  and  $Q$  intersect at most  $8n$  times, where  $|P \cup Q| = n$  (the weight of an edge is its length).

© 2004 Elsevier B.V. All rights reserved.

**Keywords:** Computational geometry; Geometric graph; Plane spanning tree; Plane spanning cycle

## 1. Introduction

Geometric graphs, that is graphs whose vertex set is a collection of points on the plane in general posi-

tion and its edges are straight line segments connecting pairs of vertices, have received a lot of attention lately. Numerous problems in which we want to draw graphs on the plane such that their vertices lie on the elements of a fixed point set have been studied. Ramsey type problems in which we want to color the edges or vertices of a geometric graph such that some specific forbidden subgraphs do not appear have also been explored. The interested reader may consult a recent survey by Pach [10] containing many results in this field.

\* Corresponding author.

E-mail addresses: [gc@cis.ibaraki.ac.jp](mailto:gc@cis.ibaraki.ac.jp) (M. Kano), [merino@matem.unam.mx](mailto:merino@matem.unam.mx) (C. Merino), [urrutia@matem.unam.mx](mailto:urrutia@matem.unam.mx) (J. Urrutia).

<sup>1</sup> Supported by PAPIIT(UNAM) of México, Proyecto IN110802.

<sup>2</sup> Supported by CONACYT of México, Proyecto 37540-A.

In this paper we are interested in embedding problems of geometric graphs on colored point sets. These problems have been considered for some time now, for example, in [1,2] the problem of embedding trees and alternating paths on bicolored point sets is solved. In [3–5] matching problems on colored point sets are studied. For two colored point sets we are interested in obtaining matchings in which every edge has its endpoints of different (or equal) color. A well-known result states that given a collection  $P_{2n}$  of  $2n$  points in general position,  $n$  blue, and  $n$  red, we can always match a blue with a red point in  $P_{2n}$  such that the line segments joining matched pairs of points do not intersect. For a recent survey dealing with numerous problems on colored point sets see [6]. Problems in which instead of coloring the vertices, we color the edges of geometric graphs, are studied in [7,8].

Let  $P_n$  be a set with  $n$  points on the plane. A spanning tree of  $P_n$  is a connected geometric graph with vertex set  $P_n$  containing exactly  $n - 1$  edges. Let  $P$  and  $Q$  be disjoint point set. Tokunaga [11] studied and solved the problem of finding spanning trees for  $P$  and  $Q$  with the smallest possible number of edge intersections. It turns out that this number depends only on the order in which the elements of  $P \cup Q$  appear in  $\text{Conv}(P \cup Q)$ . More specifically, let  $p_0, \dots, p_{r-1}$  be the points on  $\text{Conv}(P \cup Q)$  in clockwise order, and let  $i$  be the number of indexes  $j$  such that  $p_j$  and  $p_{j+1}$  are one in  $P$  and the other in  $Q$ , addition taken mod  $r$ . Then it is always possible to find spanning trees for  $P$  and  $Q$  such that their edges do not intersect if  $i \leq 2$ , otherwise they intersect exactly  $(i - 2)/2$  times. This implies, for example, that if all the points on the convex hull are the same color, or all the points in  $P$  which belong to the convex hull appear in consecutive order, then we can find spanning trees for  $P$  and  $Q$  which do not intersect, regardless of how many or where the remaining points of  $P$  and  $Q$  are.

In this paper we study the following problem: Let  $P_1, \dots, P_k$  be a family of disjoint point sets such that  $P_1 \cup \dots \cup P_k$  is in general position. Find spanning trees for  $P_1, \dots, P_k$  such that their edges have as few intersections as possible. We prove the following result:

**Theorem 1.** *Let  $P_1, \dots, P_k$  be a collection of disjoint point sets. Then we can find for each  $P_i$  a spanning tree  $T_i$  such that the total number of intersections among the edges of  $T_1, \dots, T_k$  is at most*

*$(k - 1)(n - k) + \frac{1}{2}k(k - 1)$  where  $|P_1 \cup \dots \cup P_k| = n$ . This bound is tight within a factor of two from the optimal solution.*

We also give similar results for spanning cycles of families of point sets  $P_1, \dots, P_k$  in which  $\text{Conv}(P_1) \cap \dots \cap \text{Conv}(P_k)$  is nonempty. Sharp bounds for this problem are obtained.

In the last section of this paper, we prove the following result that is of independent interest: Let  $P_1, \dots, P_k$  be families of disjoint point sets such that  $P_1 \cup \dots \cup P_k$  is in general position. For each  $P_i$  let  $T_i$  be its Euclidean minimum weight spanning tree,  $i = 1, \dots, k$ . Then the edges of these trees intersect at most  $8kn$  times.

## 2. Spanning trees with few intersections

Given two disjoint point sets  $P_1$  and  $P_2$  it is not always possible to find spanning trees for them such that their edges do not intersect. In fact if we have  $2s$  points which are the vertices of a convex polygon such that alternately they belong to  $P_1$  and  $P_2$ , then it is easy to verify that any spanning tree for  $P_1$  intersects any spanning tree for  $P_2$  at least  $s - 1$  times, see Fig. 1.

From here the following observation follows:

**Observation 1.** There are families of point sets  $P_1, \dots, P_k$  with  $|P_1 \cup \dots \cup P_k| = sk$  such that their edges intersect at least  $\frac{1}{2}k(k - 1)(s - 1)$  times.

If we consider a similar problem for three or more point sets our problem becomes much harder, even for points in convex position. Let  $P$  be a set of

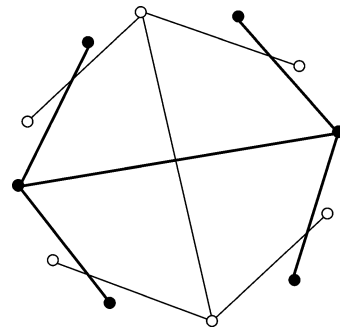


Fig. 1. Two spanning trees on 6 vertices that intersect 5 times.

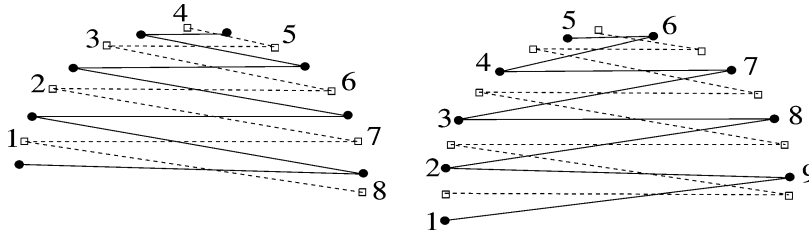


Fig. 2. On the left-hand side we have  $T_i$ ,  $i \leq \lfloor \frac{1}{2}k \rfloor$ , with solid lines and  $T_j$ ,  $j > \lfloor \frac{1}{2}k \rfloor$ , with dashed lines;  $s = 8$  so, they intersect 7 times. On the right-hand side we have  $T_i$  and  $T_j$ ,  $i, j \leq \lfloor \frac{1}{2}k \rfloor$ ; as  $s = 9$ , they intersect 15 times.

$n = ks$  points in convex position labeled  $p_1, \dots, p_{sk}$ . Split  $P$  into  $k$  subsets  $P_1, \dots, P_k$  such that the element  $p_{i+rk}$  belongs to  $P_i$ ,  $r = 0, \dots, s - 1$ . Finding for each  $P_i$  a spanning tree  $T_i$ ,  $1 \leq i \leq k$ , such that their edges have the fewest possible number of intersections is an interesting open problem. We now show a set of spanning trees  $T_1, \dots, T_k$  such that their edges intersect at most  $(\frac{3}{4}k^2 - k)(s - 1) - \frac{1}{4}k(k - 2)$  times if  $k$  is even; otherwise they intersect  $(\frac{3}{4}(k - 1)^2 + \frac{1}{2}(k - 1))(s - 1) - \frac{1}{4}(k - 1)^2$  times, i.e., the number of times their edges intersect is at most  $\frac{3}{2}$  times that of the optimal solution.

For  $i \leq \lfloor \frac{1}{2}k \rfloor$ , let  $T_i$  be the tree containing the edges joining  $p_{i+ak}$  to  $p_{i+bk}$ ,  $a + b = s + 1$  or  $a + b = s + 2$ ,  $1 \leq a, b \leq s$ . For  $i > \lfloor \frac{1}{2}k \rfloor$ ,  $T_i$  is the tree containing the edges joining  $p_{i+ak}$  to  $p_{i+bk}$ ,  $a + b = s$  or  $a + b = s + 1$ ,  $1 \leq a, b \leq s$ . Notice that two trees  $T_i$  and  $T_j$  intersect  $s - 1$  times if  $i \leq \lfloor \frac{1}{2}k \rfloor < j$ ; otherwise they intersect  $2(s - 1) - 1 = 2s - 3$  times. See Fig. 2. Therefore these trees intersect exactly  $(\frac{3}{4}k^2 - k)(s - 1) - \frac{1}{4}k(k - 2)$  if  $k$  is even; and  $(\frac{3}{4}k^2 + \frac{1}{2}(k - 1))(s - 1) - \frac{1}{4}(k - 1)^2$  if  $k$  is odd. Moreover we believe that this configuration is, in fact, close to the optimal solution for point sets in convex position. The case when  $k = 3, 4$  has been proven in [9].

We proceed now to study our problem for point sets in general position. Suppose without loss of generality that the points in  $P_1 \cup \dots \cup P_k$  have different  $x$ -coordinates, and  $|P_i| \geq 2$ ,  $i = 1, \dots, k$ . Assume that for every  $i$  the elements of  $P_i$  are labeled  $p_{i,1}, p_{i,2}, \dots, p_{i,r_i}$  such that if  $r < s$  then the  $x$ -coordinate of  $p_{i,r}$  is smaller than the  $x$ -coordinate of  $p_{i,s}$ . Let  $T_i$  be the path with vertex set  $P_i$  in which  $p_{i,j}$  is connected to  $p_{i,j+1}$  by an edge denoted by  $e_{i,j}$ ,  $j = 1, \dots, r_i - 1$ . See Fig. 3.

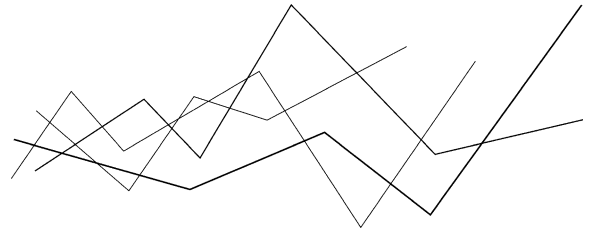


Fig. 3. The construction for Lemma 1 for 4 collections of points.

**Lemma 1.** *The edges of  $T_i$  and  $T_j$  intersect at most  $r_i + r_j - 3$  times.*

**Proof.** Our result is clearly true if  $r_i + r_j \leq 4$ , or one of  $T_i$  or  $T_j$  has exactly one edge. Suppose now that the  $x$  coordinate of  $p_{i,2}$  is smaller than that of  $p_{j,2}$ . Then the edge  $e_{i,1}$  of  $T_i$  joining  $p_{i,1}$  to  $p_{i,2}$  intersects at most one edge of  $T_j$ , namely the edge  $e_{j,1}$  joining  $p_{j,1}$  to  $p_{j,2}$ . Remove  $p_{i,1}$  from  $P_i$ , and by induction our result follows.  $\square$

In a similar way we can prove:

**Lemma 2.** *The edges of  $T_1, \dots, T_k$  intersect at most  $(k - 1)(n - k) + \frac{1}{2}k(k - 1)$  times, where  $|P_1 \cup \dots \cup P_k| = n$ .*

**Proof.** Our result is true if  $T_1, \dots, T_k$  have together at most  $k$  edges, in fact in this case if all of them intersect each other, their total number of intersections is  $\frac{1}{2}k(k - 1)$ . Suppose then that our trees contain more than  $k$  edges, and let  $e_{i,1}$  be such that the  $x$ -coordinate of  $p_{i,2}$  is smaller than the  $x$ -coordinate of  $p_{j,2}$ ,  $i \neq j$ ,  $1 \leq j \leq k$ . Then the edge  $e_{i,1}$  joining  $p_{i,1}$  to  $p_{i,2}$  intersects at most  $k - 1$  edges, i.e., in each  $T_j$ ,  $e_{i,1}$  intersects at most the edge  $e_{j,1}$  joining  $p_{j,1}$  to  $p_{j,2}$ ,  $j \neq i$ . Removing this edge, and  $p_{i,1}$  from  $P_i$ , and proceeding by

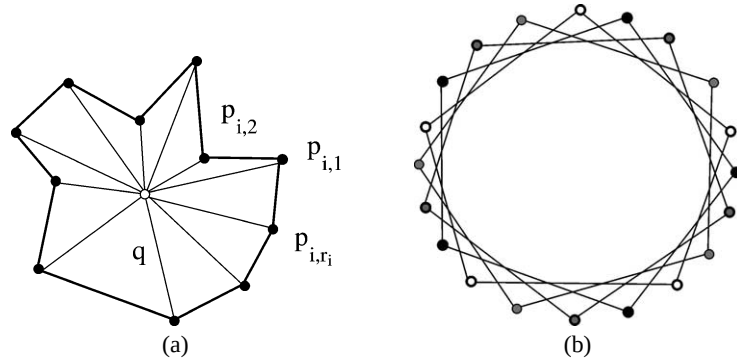


Fig. 4.

induction on  $P_1, \dots, P_i - \{p_{i,1}\}, \dots, P_k$  our result follows.  $\square$

Observe that the bound determined in Lemma 2 is within a factor of two of that in Observation 1. Theorem 1 follows directly from Observation 1 and Lemma 2.

### 3. Spanning cycles

We now study the following problem: Let  $P_1, \dots, P_k$  be a family of disjoint point sets such that  $\text{Conv}(P_1) \cap \dots \cap \text{Conv}(P_k) \neq \emptyset$ . Find a family of spanning cycles  $C_1, \dots, C_k$  of  $P_1, \dots, P_k$ , respectively, with few intersections. We prove:

**Theorem 2.** *Let  $P_1, \dots, P_k$  be disjoint point sets such that  $\text{Conv}(P_1) \cap \dots \cap \text{Conv}(P_k) \neq \emptyset$ . Then for each  $P_i$  we can find a cycle  $C_i$  which covers the vertices of  $P_i$  such that the edges of all cycles  $C_1, \dots, C_k$  intersect at most  $(k-1)n$  times. Our bound is optimal.*

**Proof.** Let  $q$  be a point in the interior of  $\text{Conv}(P_1) \cap \dots \cap \text{Conv}(P_k)$ . For each  $P_i$  define a cycle  $C_i^q$  as follows: Sort the elements of  $P_i$  around  $q$  in the counterclockwise order according to their slope and label them  $p_{i,1}, \dots, p_{i,r_i}$ , see Fig. 4(a).

A straightforward modification to our counting argument in Lemma 2 shows that the edges of  $C_1, \dots, C_k$  intersect at most  $(k-1)n$  times. To show that our bound is tight, choose  $n = kr$ , and choose  $kr$  points on a unit circle labeled  $p_1, \dots, p_{kr}$ , and let  $P_i = \{p_{i+k_s} : k = 0, \dots, r-1\}$ . It is easy to see that the

(unique) cycles  $C_i$  that cover the vertices of each  $P_i$ ,  $i = 1, \dots, k$  intersect  $(k-1)kr = (k-1)n$  times, see Fig. 4(b).  $\square$

### 4. Minimum weight spanning trees

The euclidean minimum weight spanning tree of a point set  $P_n$  is a tree with vertex set  $P_n$  such that the sum of the lengths of its edges is minimized. In this section we prove that if  $P_1, \dots, P_k$  are disjoint point sets and  $T_1, \dots, T_k$  are their corresponding euclidean minimum weight spanning trees then the total number of intersections among their edges is at most  $8(k-1)(n-k)$  where  $|P_1 \cup \dots \cup P_k| = n$ . Our proof is based on the following observation that is easy to prove: Let  $T_i$  and  $T_j$  be the minimum weight spanning trees of  $P_i$  and  $P_j$ . Let  $e$  be any edge of  $T_i$ . Then there is a constant  $c$  such that  $e$  intersects at most  $c$  edges of  $T_j$  whose length is greater than or equal to the length of  $e$ . It follows that the edges of  $T_i$  and  $T_j$  intersect a linear number of times. In fact, we can prove that  $c$  is at most 8, however the proof is long, tedious, and unenlightening. We skip the details, they can be supplied by the authors upon request. Summarizing we have:

**Lemma 3.** *Let  $T_1$  and  $T_2$  be the minimum weight spanning trees of two point sets  $P_1$  and  $P_2$ , and  $e$  any edge of  $T_1$ . Then  $e$  intersects at most 8 edges of  $T_2$  whose length is greater than or equal to the length of  $e$ .*

We now prove:

**Theorem 3.** *Let  $T_1, \dots, T_k$  be respectively the minimum weight spanning trees of  $k$  point sets  $P_1, \dots, P_k$*

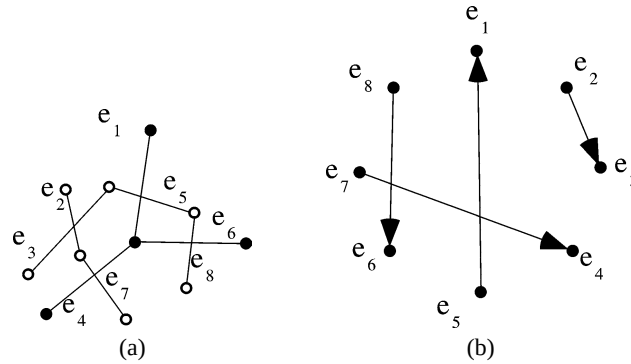


Fig. 5.

such that  $|P_1 \cup \dots \cup P_k| = n$ . Then the edges of  $T_1, \dots, T_k$  intersect at most  $8(k-1)(n-k)$  times.

**Proof.** Observe first that since  $|P_1 \cup \dots \cup P_k| = n$ ,  $T_1, \dots, T_k$  have exactly  $n - k$  edges. Let us construct the intersection graph  $H$  of the set of edges of  $T_1, \dots, T_k$ , that is the graph whose vertex set is the set of all edges of  $T_1, \dots, T_k$ , two of which are adjacent if they intersect. Orient the edges of this graph as follows: If two edges  $e \in T_i$  and  $e' \in T_j$  intersect and  $e$  is longer than  $e'$  orient the edge in  $H$  joining them from  $e$  to  $e'$ , see Fig. 5.

By Lemma 3 every edge in  $T_i$  intersects at most 8 edges in each  $T_j$ ,  $i \neq j$  which are the same length or longer than itself. Thus the out-degree of each vertex of  $H$  is at most  $8(k-1)$ . Our result follows.  $\square$

Observe that for the case when we have two point sets  $P$  and  $Q$  such that  $|P \cup Q| = n$ , our previous results implies that the edges of their minimum weight spanning trees intersect at most  $8(n-2)$  times. This bound is far from optimal. In fact we have been unable to produce examples in which the minimum weight spanning trees of  $P$  and  $Q$  intersect more than  $2n-4$  times. An example is constructed as follows:  $P$  consists of 3 points  $r, s, t$  such that  $r$  and  $s$  are equidistant from  $t$ , and the angle  $\angle rts$  is slightly bigger than  $\pi/3$ . The points of  $Q$  lie on a zig-zag polygonal such that each second segment of it is parallel, and the angle between two consecutive segments is  $\pi/2$  as shown in Fig. 6.

We conclude by posing the following question:

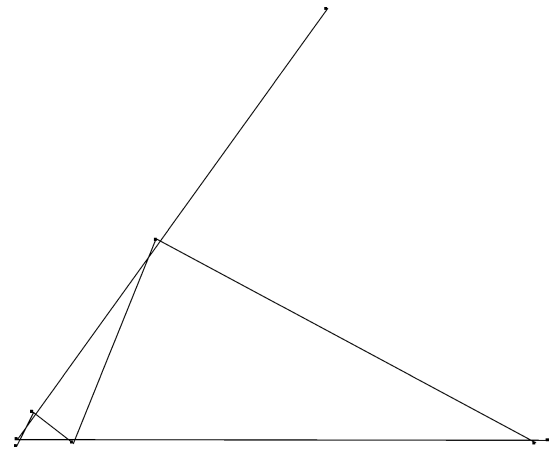


Fig. 6.

**Open problem 1.** Is it true that the edges of the minimum weight spanning trees of any two point sets  $P$  and  $Q$  such that  $|P \cup Q| = n$  intersect at most  $2n - c$  times,  $c$  a constant?

## References

- [1] M. Abellanas, J. García, G. Hernández, N. Noy, P. Ramos, Bipartite embeddings of trees in the plane, in: Graph Drawing 96, in: Lecture Notes in Comput. Sci., vol. 1990, Springer-Verlag, Berlin, 1997, pp. 1–10.
- [2] J. Akiyama, J. Urrutia, Simple alternating path problems, Discrete Math. 84 (1990) 101–103.
- [3] A. Dumitrescu, J. Pach, Partitioning colored point sets into monochromatic parts, Int. J. Comput. Geom. Appl. 12 (5) (2002) 401–412.
- [4] A. Dumitrescu, R. Kaye, Matching colored points in the plane, some new result, Comput. Geom., Theory Appl. 19 (1) (2001) 69–85.

- [5] A. Dumitrescu, W. Steiger, On a matching problem on the plane, *Discrete Math.* 211 (2000) 183–195.
- [6] A. Kaneko, M. Kano, Some problems and results on graph theory and discrete geometry, 2001; ps-file in <http://gorogoro.cis.ibaraki.ac.jp/>.
- [7] G. Karoli, J. Pach, G. Tóth, Ramsey-type results for geometric graphs, I, *Discrete Comput. Geom.* 18 (1995) 247–255.
- [8] G. Karoli, J. Pach, G. Tóth, P. Valtr, Ramsey-type results for geometric graphs, II, *Discrete Comput. Geom.* 20 (1998) 375–388.
- [9] J. Leaños, C. Merino, G. Salazar, J. Urrutia, Spanning trees of multicoloured point sets with few intersections, Presented at the JCDCG 2003, Indonesia.
- [10] J. Pach, Geometric graph theory, in: *Surveys in Combinatorics, 1999 (Canterbury)*, in: *London Math. Soc. Lecture Note Ser.*, vol. 267, Cambridge Univ. Press, Cambridge, 1999, pp. 167–200.
- [11] S. Tokunaga, Crossing number of two-connected geometric graphs, *Inform. Process. Lett.* 59 (1996) 331–333.